

MONT-DE-MARSAN AB, France

WMO: 076070

Lat: 43.9097N

Lon: 0.5003W

Elev: 62

StdP: 100.58

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -4.9	(c) -3.3	(d) -7.4	(e) 2.0	(f) -1.9	(g) -5.4	(h) 2.4	(i) -0.7	(j) 9.5	(k) 10.6	(l) 8.2	(m) 10.7	(n) 1.1	(o) 80	(p) 0.446

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a) 7	(b) 12.2	(c) 33.4	(d) 22.0	(e) 31.3	(f) 21.1	(g) 29.5	(h) 20.4	(i) 23.2	(j) 30.6	(k) 22.2	(l) 29.3	(m) 21.4	(n) 27.6	(o) 3.0	(p) 120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a) 21.0	(b) 15.7	(c) 26.4	(d) 20.0	(e) 14.8	(f) 24.8	(g) 19.2	(h) 14.1	(i) 23.9	(j) 69.0	(k) 30.6	(l) 65.4	(m) 29.3	(n) 62.2	(o) 27.9	(p) 27.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7.9	6.6	5.5	-8.0	37.3	2.3	1.9	-9.7	38.7	-11.0	39.8	-12.3	40.8	-13.9	42.2
			-8.3	25.1	2.3	1.1	-9.9	25.9	-11.3	26.5	-12.5	27.1	-14.2	27.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.6	6.3	6.9	10.0	12.4	15.9	19.4	21.3	21.3	18.1	14.8	9.3	6.7		
	DBStd	6.48	3.77	3.72	3.44	3.31	3.45	3.55	3.16	3.23	3.24	3.69	3.64	4.03		
	HDD10.0	449	123	95	43	13	2	0	0	0	0	7	54	113		
	HDD18.3	2061	374	319	258	178	92	29	9	43	118	270	360			
	CDD10.0	1752	8	9	43	86	184	281	349	349	242	156	34	11		
	CDD18.3	321	0	0	0	1	16	60	100	99	36	9	0	0		
	CDH23.3	3359	0	1	8	56	223	648	968	973	403	78	1	0		
	CDH26.7	1235	0	0	0	8	60	246	383	398	126	14	0	0		
(14) Wind	WSAvg	2.4	2.3	2.4	2.6	2.8	2.7	2.6	2.5	2.3	2.0	2.0	2.1	2.1		
(15) Precipitation	PrecAvg	969	96	73	70	93	82	68	56	60	75	90	122	83		
	PrecMax	1230	222	137	151	230	195	175	126	175	152	194	313	172		
	PrecMin	764	16	7	11	26	27	9	12	12	22	27	26	6		
	PrecStd	127	56	41	41	50	36	38	28	37	36	49	69	41		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.3	20.6	24.5	27.7	31.2	35.3	35.5	37.3	32.5	28.6	21.3	17.9		
		MCWB	12.6	12.6	14.0	16.8	20.0	22.7	22.4	22.6	20.6	19.0	15.7	12.6		
	2%	DB	15.0	17.4	21.7	24.9	28.2	31.7	32.9	33.4	29.9	25.4	18.8	15.5		
		MCWB	11.6	11.5	13.4	15.3	18.4	21.1	21.9	22.3	20.0	18.4	14.4	11.9		
	5%	DB	13.3	15.1	19.3	22.3	25.8	29.5	30.8	30.8	27.9	23.2	16.6	13.9		
		MCWB	11.0	10.6	12.1	14.1	17.6	20.3	21.1	21.3	19.2	17.3	13.6	11.5		
	10%	DB	12.0	13.0	17.0	19.8	23.3	27.2	28.7	28.5	25.6	21.1	14.9	12.5		
		MCWB	10.2	10.1	11.3	13.0	16.4	19.4	20.5	20.5	18.3	16.5	12.7	10.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.6	13.6	15.2	17.8	21.0	24.0	24.2	24.6	22.7	20.3	16.7	14.0		
		MCDB	15.3	18.0	21.3	25.4	28.9	31.6	31.8	32.7	29.4	25.3	19.5	15.7		
	2%	WB	12.5	12.3	14.1	16.2	19.4	22.4	22.9	23.3	21.1	19.2	15.2	12.8		
		MCDB	14.2	15.4	19.6	23.1	26.7	29.9	30.8	30.4	27.1	23.6	17.6	14.6		
	5%	WB	11.5	11.4	13.0	14.9	18.2	21.1	22.0	22.3	20.1	18.2	14.0	11.9		
		MCDB	12.9	14.0	17.9	20.4	24.2	28.0	29.3	28.9	25.5	21.8	15.9	13.4		
	10%	WB	10.4	10.5	12.1	13.7	17.0	20.0	21.0	21.3	19.1	17.2	13.0	10.9		
		MCDB	11.7	12.8	15.8	18.3	22.1	26.0	27.1	27.0	24.0	20.3	14.7	12.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	10.6	12.0	11.9	11.8	12.1	12.2	12.8	13.1	11.6	9.2	8.8		
		MCDBR	10.6	14.8	16.7	17.0	16.4	16.6	16.6	17.3	17.2	14.5	11.7	10.5		
		MCWBR	7.4	9.3	8.9	8.5	7.7	7.4	6.8	7.2	7.9	7.7	7.7	7.2		
		MCDBR	8.7	12.1	13.5	14.5	14.2	14.6	14.3	14.4	14.2	12.1	10.1	8.7		
	5% WB	MCWBR	6.5	8.0	7.5	7.5	7.1	7.1	6.2	6.3	6.9	6.7	7.1	6.4		
(40) Clear-Sky Solar Irradiance	taub		0.325	0.340	0.373	0.399	0.407	0.420	0.411	0.409	0.382	0.365	0.341	0.322		
	taud		2.489	2.428	2.342	2.275	2.304	2.300	2.327	2.343	2.404	2.457	2.472	2.488		
	Ebn at Noon		801	850	861	864	866	853	856	845	842	809	771	766		
	Edh at Noon		71	89	110	127	127	128	123	118	102	86	72	65		
(44) All-Sky Solar Radiation	RadAvg		1.42	2.34	3.51	4.50	5.22	5.87	5.84	5.23	4.26	2.78	1.68	1.32		
	RadStd		0.13	0.28	0.40	0.48	0.44	0.42	0.41	0.35	0.23	0.23	0.20	0.15		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (20 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page